

Deriving Privacy Settings for Location Sharing: Are Context Factors Always the Best Choice?

Rishindra Mateti

The paper, authored by Frederic Raber and Antonio Krueger, investigates whether context factors alone are sufficient for determining privacy settings in location-sharing scenarios or if individual traits, such as personality and privacy concerns, play a critical role. The study aims to predict fine-grained privacy preferences across seven levels of location disclosure, ranging from exact GPS coordinates to complete obfuscation. The authors argue that a model incorporating both context and individual factors achieves a prediction accuracy that is 20% better than a constant baseline. This research challenges the conventional approach of binary privacy settings and highlights the need for adaptive, user-specific privacy solutions.

The authors provide a clear explanation of key terms such as privacy levels, location obfuscation, and personality measures, ensuring a well-defined research scope. The paper is well-structured and logically organized, presenting background research, methodology, results, and conclusions in a coherent manner. The writing is clear and concise, with technical concepts effectively communicated to the audience. Furthermore, the inclusion of statistical analysis and machine learning techniques strengthens the credibility of the findings. The paper adheres to academic writing standards, maintaining clarity with minimal grammatical or spelling errors.

The main findings of the study indicate that both personality traits and privacy concerns significantly influence location-sharing decisions. The study's user survey revealed that individuals high in openness and extraversion are more inclined to share precise location data, whereas those with strong privacy concerns prefer higher levels of obfuscation. Additionally, the research confirms that users favor fine-grained privacy settings, with "City Only" being the most frequently selected level. This suggests that binary privacy choices (full disclosure or complete restriction) are inadequate for addressing users' real-world privacy needs.

The paper's primary strength is its empirical approach, combining user studies with regression analysis to validate its predictive model. A survey with 100 participants measured privacy preferences across seven location-sharing levels, with "City Only" being the most frequently chosen option (32.65%), followed by "No Location" (22.53%) and "Exact Location" (17.84%) (page 89, Table I). The study applied Categorical Regression (CATREG), achieving an apparent prediction error (APE) of 0.808, demonstrating a 20% improvement over a constant baseline (page 91, Table III). The model's accuracy improved when integrating IUIPC privacy measures, with the strongest correlations found for privacy control (0.106, $p < 0.001$) and collection (0.167, $p < 0.001$) (page 90, Table II). Additionally, openness (-0.071, $p < 0.001$) and extraversion (-0.068, $p < 0.001$) correlated with higher disclosure, while conscientiousness (0.060, $p < 0.001$) led to increased obfuscation. These results highlight the importance of combining context and individual factors in privacy decision-making.

One weakness of the paper is its reliance on self-reported survey data, which may not align with real-world behavior due to the privacy paradox (page 89, citing [1]). Users often express privacy concerns but disclose more than they claim, particularly in social or convenience-driven situations (page 88). The study sample, though diverse in age (19–65, $M=33.08$, $SD=9.14$), was limited to 100 participants from Prolific Academic, raising concerns about demographic representation (page 89). The regression model explained 19.3% of variance in privacy decisions, indicating additional factors may be influential (page 91, Table III). The authors acknowledge this limitation and suggest future studies validate the model using real-world behavioral data from platforms like Facebook or Google Maps (page 92). Expanding the dataset would improve the model's generalizability and practical application.

Future studies should validate the model with a larger participant base and explore how privacy preferences change based on situational factors like proximity and social context. Real-world data collection from platforms like Facebook or Google Maps would improve practical applicability. Testing user interactions with automated privacy recommendations could further refine the balance between user control and system-generated suggestions.

In conclusion, the paper provides valuable insights into the role of personality and privacy concerns in location-sharing decisions. It demonstrates that context factors alone are not always the best predictors and that integrating user-specific traits improves privacy-setting recommendations. While the study presents a solid foundation, further research is needed to validate findings in real-world settings and explore user interactions with automated

privacy tools. Overall, the paper contributes meaningfully to privacy-aware computing and offers practical implications for future location-sharing systems.